Betweenness Centrality Prediction with Hyper-Connected GNNs

Project Progress Report

Research Log Synthesis

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Contents

1 Research Question

Can deep GNNs predict betweenness centrality (BC) inductively, and do HC / mHC / mHC-lite improve depth scaling and out-of-distribution (OOD) ranking quality compared with strong deep-GNN baselines?

Primary evaluation metrics:

- Ranking: Kendall τ, Spearman ρ
- Top-K: Precision@K, NDCG@K

Protocol (inductive): train on small synthetic graphs (100 nodes, ER/BA/SBM), evaluate on larger unseen graphs (500 nodes, same families = ID; WS/RGG families = OOD).

2 Stage 1 — Full 5-Seed Depth Sweep (2026-03-06)

2.1 Setup

Seeds	5 (0–4)
Models	19 variants: GCN, SAGE, GAT, GIN, GCNII, APPNP, JKNet, HC, mHC, mHC-lite
Depths	$L \in \{2, 4, 8, 16, 32\}$
Train graphs	160 (100 nodes, ER/BA/SBM)
Val / Test ID / OOD	30 / 30 / 30 (500 nodes)
Budget	50 epochs, patience 10, LR 10^{-3}, WD 10^{-5}

2.2 Best points from the frozen sweep

Category	Model	L	OOD Kendall
Best overall OOD	`hc_gcnii`	2	0.3824 ± 0.0253
Best mHC point	`mhc_gcn`	2	0.3812 ± 0.0238
Strongest GCN gain	`hc_gcn`	4	0.3809 ± 0.0386
Best deep baseline	`appnp`	32	0.3544 ± 0.0258

2.3 Conclusions

HC / mHC improve OOD consistently; gains concentrate at $L \leq 8$. No ultra-deep scaling claim is supported. Two families retained: **GCN** (clearest mHC signal) and **GCNII** (deep-backbone stress test).

3 Stage 2 — Targeted Hyperparameter Search (2026-03-06)

Optuna proxy search (15 trials, validation-only objective):

- **GCN Trial 3** (obj. 0.8605): LR 2.26×10^{-4}, WD 2.10×10^{-5}, dropout 0.1, $n_{\text{str}} = 2$, $\tau = 0.2$, iters 20.
- **GCNII Trial 11** (obj. 0.8564): LR 1.07×10^{-3}, WD 4.34×10^{-5}, $n_{\text{str}} = 4$, $\tau = 0.1$, GCNII $\alpha = 0.1$, $\theta = 0.5$.

GCNII failure at L16 persists after tuning. GCN + mHC is the strongest candidate.

4 Stage 3 — Multi-Seed Confirmation (2026-03-07)

90 runs per family (3 candidate trials × 3 depths × 5 seeds).

GCN: mHC-GCN consistently gains on OOD Kendall at L4/L8 (strongest: +0.071 at t3/L4). Single negative case: t13/L2 (−0.006).

GCNII: mHC helps at L2/L8 but degrades at L16 (deltas: −0.039, −0.012, −0.047).

Decision: GCN family only from this point.

5 Stage 4 — GCN Ablation Study (2026-03-07)

5.1 Results at L8 (tuned GCN regime, 5 seeds)

Block	Variant	Val Kendall	OOD Kendall
Structure	gcn_ref	0.8332	0.3574
Structure	mhc_gcn_ref	0.8909	0.3888
Structure	**hc_gcn_ref**	**0.9174**	**0.4054**
Structure	dynamic only	0.8882	0.3904
Structure	static only	0.8832	0.3855
Structure	$n = 4$ streams	0.8809	0.3791
Sinkhorn	$\tau = 0.05$	0.8873	0.3923
Sinkhorn	$\tau = 0.10$	0.8932	0.3973
Sinkhorn	$\tau = 0.20$	0.8952	0.3911
Sinkhorn	iters = 10	0.8886	0.3906

5.2 Interpretation

HC-GCN dominates. No mHC variant exceeds HC regardless of τ tuning. $n = 2$ streams is optimal. Dynamic and static components contribute equally. The useful mechanism is flexible multi-stream routing, *not* the doubly-stochastic Sinkhorn constraint.

6 Stage 5 — Final GCN Family Comparison (2026-03-09)

Table 1: Final GCN family — test OOD results (5 seeds)

Variant	L	Kendall	Spearman	P@10	NDCG@10
gcn	4	0.302 ± 0.045	0.423 ± 0.063	0.247	0.513
gcn	8	0.360 ± 0.021	0.506 ± 0.028	0.264	0.563
hc_gcn	4	0.393 ± 0.031	0.543 ± 0.041	**0.291**	**0.571**
hc_gcn	8	$\mathbf{0.406 \pm 0.041}$	$\mathbf{0.558 \pm 0.054}$	0.287	0.567
mhc_gcn	4	0.374 ± 0.043	0.521 ± 0.055	0.274	0.566
mhc_gcn	8	0.396 ± 0.038	0.548 ± 0.049	0.278	**0.568**
mhc_lite_gcn	4	0.376 ± 0.042	0.523 ± 0.055	0.271	0.553
mhc_lite_gcn	8	0.395 ± 0.039	0.544 ± 0.050	0.282	0.561

Key findings:

1. **Val:** HC gains are clearly learned in validation and remain visible on Kendall, Precision@10, and NDCG@10. At $L = 8$, HC-GCN reaches 0.918 Kendall, 0.919 Precision@10, and 0.995 NDCG@10, versus 0.833, 0.851, and 0.976 for plain GCN. Depth has little effect on plain GCN, while mHC and mHC-lite improve from $L = 4$ to $L = 8$.
2. **Test ID:** All models remain in a narrow band on in-distribution test Kendall (roughly 0.82–0.83), suggesting that hyper-connections bring little benefit on the known distribution.
3. **Test OOD:** Gains are substantial on out-of-distribution graphs: HC-GCN > mHC-GCN ≈ mHC-lite-GCN > GCN on the main ranking metrics. On top-K at 10, HC is strongest on Precision@10, while mHC is marginally strongest on NDCG@10.

7 Stage 6 — Residual Baseline (2026-03-10)

Table 2: Residual baseline — OOD Kendall (5 seeds)

Variant	L	OOD Kendall	Relative gain
gcn	4	0.305 ± 0.046	—
gcn_residual	4	0.352 ± 0.033	+15.2%
hc_gcn	4	0.395 ± 0.037	+29.3%
mhc_gcn	4	0.380 ± 0.049	+24.4%
gcn	8	0.359 ± 0.024	—
gcn_residual	8	0.369 ± 0.035	+2.6%
hc_gcn	8	$\mathbf{0.411 \pm 0.040}$	+14.5%
mhc_gcn	8	0.389 ± 0.046	+8.4%

At L8: residual gives only +2.6%; HC-GCN gives +14.5%. The hyper-connection gain is architectural, not reducible to skip connections.

8 Stage 7 — Zero-Shot Real-Graph Transfer (2026-03-09/10)

8.1 Setup

Models trained only on synthetic graphs evaluated zero-shot on:

- **Cora** (2708 nodes, 5278 edges, $\bar{d} \approx 3.9$)
- **CiteSeer** (3327 nodes, 4552 edges, $\bar{d} \approx 2.7$)
- **PubMed** (19717 nodes, 44324 edges, $\bar{d} \approx 4.5$)

8.2 Results — Kendall τ (mean ± std, 5 seeds)

Variant	Cora	CiteSeer	PubMed
gcn_l8	0.350 ± 0.053	0.067 ± 0.118	0.539 ± 0.025
gcn_residual_l8	0.355 ± 0.062	0.069 ± 0.080	0.605 ± 0.042
hc_gcn_l8	0.366 ± 0.034	0.056 ± 0.029	0.595 ± 0.028
mhc_gcn_l8	$\mathbf{0.405 \pm 0.036}$	$\mathbf{0.151 \pm 0.083}$	0.598 ± 0.030
mhc_lite_gcn_l8	0.380 ± 0.037	0.125 ± 0.085	$\mathbf{0.611 \pm 0.030}$

8.3 Per-graph interpretation

Cora ($\bar{d} \approx 3.9$). All hyper-connected variants improve. BC is not severely skewed. mHC-GCN leads on all metrics (+15.5% Kendall, +43.8% P@top5%). HC also gains (+4.5%) without regression.

CiteSeer ($\bar{d} \approx 2.7$ — extreme sparsity). Most informative graph. BC concentrates on a small number of bridge nodes. HC-GCN *regresses* −16% in Kendall vs. plain GCN, while achieving best NDCG@10 (+34%): HC learns to route information toward bridges (correct top-K) but destroys global ranking of other nodes. The Sinkhorn constraint of mHC prevents this collapse: +127% Kendall gain. The routing matrix analysis (Section **??**) provides a mechanistic account of this divergence.

PubMed (19717 nodes). Gains more moderate. GCN+residual is very competitive (+12.2%). mHC-lite leads (+13.3%): full Sinkhorn overhead not worthwhile at this scale; lighter variant offers better trade-off. Precision@10 saturates at 0.5 for all models (non-discriminative).

8.4 Routing collapse on sparse BC

The HC regression on CiteSeer reveals a regime-dependent property:

HC (unconstrained)	**mHC (Sinkhorn)**
Best on synthetic BC (moderate skew)	Best on real citation graphs
High routing freedom	Proportional stream balancing
Collapses on sparse/skewed BC	No regression on any tested graph

9 Stage 8 — Routing Matrix Analysis (2026-03-11)

9.1 Setup

Post-hoc analysis of the inter-stream residual matrices H_{res} on three models across three regimes (synthetic ID, synthetic OOD, Cora), 5 seeds.

Models	`hc_gcn_l8`, `mhc_gcn_l8`, `mhc_lite_gcn_l8`
Regimes	synthetic ID, synthetic OOD, Cora
Seeds	5
Layers	all 8
Statistics	identity distance, row entropy, row/col sum error, permutation proximity

9.2 HC-GCN: progressive, unconstrained routing freedom

H_{res} drifts progressively away from the identity as depth increases (measured on synthetic OOD):

Layer	Identity distance	Row entropy
0	≈ 0.064	≈ 0.18
3	≈ 0.182	—
7	≈ 0.951	≈ 0.75

Row and column sum errors become large in deeper layers, confirming that H_{res} is not doubly stochastic: the model is free to route probability mass unevenly across streams. This aggressive, unconstrained remixing is consistent with HC-GCN's superiority on the synthetic inductive benchmark, and explains the routing collapse on CiteSeer (Section **??**): deep layers over-commit to bridge-node streams, distorting global rankings.

9.3 mHC-GCN: strongly mixed but Sinkhorn-balanced

The raw residual matrix is already far from the identity before Sinkhorn projection (layer 7: raw identity distance ≈ 0.816): the model *wants* to mix streams aggressively. After Sinkhorn projection:

- identity distance at layer 7 ≈ 0.403 (synthetic OOD)
- row and column sum errors ≈ 0 (doubly-stochastic constraint enforced)
- the pattern is stable across synthetic ID, synthetic OOD, and Cora

The projection forces the mixing into a balanced transport plan, preserving more global structure than HC. This is consistent with mHC's stronger zero-shot transfer and the absence of regression across all tested real graphs.

9.4 mHC-lite-GCN: near-uniform, depth-invariant mixer

With $n_{\text{str}} = 2$, the routing behaves like a near-uniform convex combination of identity and swap, almost constant across layers and regimes:

- identity distance ≈ 0.707 (constant across layers)
- row entropy ≈ 1.0 (maximum)
- max permutation coefficient ≈ 0.50
- permutation entropy ≈ 1.0

The mechanism is only weakly input-adaptive, which is consistent with its stable but moderate performance on large real graphs such as PubMed.

9.5 Mechanistic summary

Table 3: Routing characterisation from learned H_{res} matrices

Variant	Routing character	Consequence
HC	Free; grows more aggressive at depth	Strongest synthetic BC; collapses on sparse/skewed graphs
mHC	Strongly mixed, Sinkhorn-balanced	Best zero-shot transfer; no regression on any graph
mHC-lite	Near-uniform, depth-invariant	Stable and competitive on large graphs (PubMed)

The synthetic vs. real ranking inversion is not a metric artefact: it has a clear mechanistic signature in the routing matrices. HC and mHC learn structurally different transport plans, and those differences map directly onto their respective failure modes.

10 Medium-Term Conclusions

10.1 What the project has established

1. **Hyper-connections improve inductive BC ranking.** Every HC-family variant outperforms GCN on OOD synthetic and on real-graph zero-shot transfer. The gain is architectural, not reducible to residual connections.
2. **The gain is OOD-specific.** On in-distribution test graphs all models hit the same ceiling. Hyper-connections transfer knowledge; they do not add raw capacity on known distributions.
3. **HC-GCN is the best variant on the synthetic protocol.**

$$\text{HC-GCN } (0.406) > \text{mHC-GCN } (0.396) > \text{mHC-lite-GCN } (0.395) > \text{GCN } (0.360)$$

4. **mHC-GCN is the most robust variant in zero-shot transfer.** It never regresses below the GCN baseline across any benchmark. HC-GCN collapses on very sparse, skewed-BC graphs (CiteSeer). The Sinkhorn constraint acts as routing regularization.
5. **Deep scaling is not validated.** GCNII + mHC fails at L16. HC-family advantages concentrate at $L \leq 8$ for GCN backbones.
6. **The performance split has a mechanistic explanation.** Matrix analysis of H_{res} shows that HC learns increasingly free, non-doubly-stochastic routing (identity distance $0.064 \rightarrow 0.951$ across layers), mHC is equally aggressive before projection but Sinkhorn-balanced after, and mHC-lite is an almost depth-invariant near-uniform mixer. The synthetic vs. real graph ranking inversion is directly visible in the routing statistics.

10.2 What cannot be claimed yet

- mHC does not universally outperform HC.
- Deep scaling ($L > 8$) is not validated for any variant.
- The HC routing collapse is a single observation (CiteSeer); additional sparse graphs are needed to generalize it.
- No parameter-count-matched comparison exists; gains could partially reflect extra capacity.

11 Stage 8 — Routing Matrix Analysis (2026-03-11)

11.1 Motivation

Performance alone tells us *which* model wins, but not *why*. To explain the synthetic vs. real-graph ranking inversion (HC best on synthetic, mHC/mHC-lite best on citation graphs), we inspected the learned inter-stream residual routing matrices H_{res}.

11.2 Setup

11.3 Matrix Statistics

For each learned routing matrix $H_{\text{res}} \in \mathbb{R}^{n_{\text{str}} \times n_{\text{str}}}$, we report the following quantities.

Models	`hc_gcn_l8`, `mhc_gcn_l8`, `mhc_lite_gcn_l8`
Regimes	synthetic ID, synthetic OOD, Cora
Seeds	5
Depth analyzed	all 8 layers
Synthetic sample	3 ID graphs + 3 OOD graphs per seed
Outputs	`outputs/analysis/hc_matrices_gcn_l8/`

Distance to identity. This measures how far the routing departs from "no mixing":

$$d_I = \|H_{\text{res}} - I\|_F$$

Small values mean that each stream mostly keeps its own information. Large values mean strong cross-stream remixing.

Row entropy. For row i, with weights w_{ij}:

$$H_i = -\sum_j w_{ij} \log w_{ij}$$

Low entropy means concentrated routing from one source stream. High entropy means diffuse mixing from several source streams.

Row/column sum error. To test whether routing is approximately doubly stochastic, we measure deviations from:

$$\sum_j H_{ij} = 1, \qquad \sum_i H_{ij} = 1$$

Small error means balanced transport; large error means unconstrained routing.

Permutation proximity. For double-stochastic routing, we also inspect proximity to permutation-like transport using a Birkhoff-style decomposition:

$$H_{\text{res}} = \sum_k \lambda_k P_k, \qquad \lambda_k \geq 0, \qquad \sum_k \lambda_k = 1$$

where P_k are permutation matrices. We track:

- `max_permutation_coeff`: dominant permutation weight
- `permutation_coeff_entropy`: concentration vs. diffusion of λ_k

11.4 HC-GCN: progressively freer routing

Table 4: Representative HC-GCN routing statistics on synthetic OOD

Layer	Identity distance	Row entropy
0	≈ 0.064	≈ 0.18
3	≈ 0.182	—
7	≈ 0.951	≈ 0.75

Observed behavior.

- H_{res} moves steadily away from the identity as depth increases.

- Row entropy also grows with depth.
- Row/column sum errors become large in deeper layers.

Interpretation. HC learns increasingly free, non-constrained inter-stream remixing. The routing becomes both stronger and more diffuse with depth. This is consistent with HC-GCN being the strongest variant on the synthetic benchmark: the task benefits from aggressive multi-stream recombination of weak structural signals.

11.5 mHC-GCN: strongly mixed but Sinkhorn-balanced

Observed behavior.

- Before Sinkhorn, the raw residual matrix already wants to mix strongly:

$$\texttt{raw_identity_distance} \approx 0.816 \quad \text{at layer 7}$$

- After Sinkhorn, the effective routing is much more controlled:

$$\texttt{identity_distance} \approx 0.403 \quad \text{at layer 7 on synthetic OOD}$$

- Row and column sum errors are essentially zero.
- This pattern is stable across synthetic ID, synthetic OOD, and Cora.

Interpretation. mHC does *not* learn weak mixing. Instead, it learns strong mixing that is projected into a balanced transport plan. The Sinkhorn constraint preserves more structure than HC and prevents unbalanced routing collapse. This matches mHC's better real-graph transfer behavior.

11.6 mHC-lite-GCN: near-uniform default mixer

Typical values across layers and regimes.

- `identity_distance` ≈ 0.707
- `row_abs_entropy` ≈ 1.0
- `max_permutation_coeff` ≈ 0.50
- `permutation_coeff_entropy` ≈ 1.0

Interpretation. With $n_{\text{str}} = 2$, mHC-lite behaves like a near-uniform convex combination of identity and swap:

$$H_{\text{res}} \approx 0.5\, I + 0.5\, P_{\text{swap}}$$

The mechanism is highly regularized, weakly input-adaptive, and nearly invariant with depth. This explains why it is stable and competitive on large real graphs such as PubMed, without being the strongest model on the synthetic benchmark.

11.7 Mechanistic Summary

Main conclusion. The synthetic vs. real-graph ranking inversion is not a metric artefact. It has a direct mechanistic signature in the routing matrices:

- synthetic inductive BC benefits from freer multi-stream routing,
- citation-graph transfer benefits more from constrained or highly regularized routing.

Table 5: Routing interpretation from learned H_{res} matrices

Variant	Routing character	Consequence
HC	Free, increasingly aggressive with depth	Best synthetic OOD ranking; strongest remixing of structural signals
mHC	Strongly mixed, but Sinkhorn-balanced	Better transfer stability; constrained transport prevents routing drift
mHC-lite	Near-uniform, almost depth-invariant mixer	Stable, highly regularized behavior; good robustness on large real graphs

12 Stage 9 — EdgeConv Family Benchmark at L8 (2026-03-11)

12.1 Motivation

The GCN-family study shows that hyper-connections are useful on a simple backbone. The next question is whether they still matter once the base convolution is already stronger. The EdgeConv benchmark separates two effects:

1. backbone strength (`EdgeConv` vs. `GCN`)
2. routing mechanism (residual vs. HC/mHC/mHC-lite)

12.2 Setup

Depth	$L = 8$
Seeds	5
New models	`edgeconv`, `edgeconv_residual`, `hc_edgeconv`, `mhc_edgeconv`, `mhc_lite_edgeconv`
Training regime	matched to final GCN-family setup
Reference family	frozen GCN-family results at $L8$
Output root	`outputs/followups/edgeconv_vs_gcn_family_l8/`

12.3 Main Results

Table 6: EdgeConv family benchmark at L8

Model	OOD Kendall	OOD Spearman	Precision@10	NDCG@10
`edgeconv_l8`	0.3321	0.4698	0.2420	0.5276
`edgeconv_residual_l8`	0.4324	0.5955	0.2613	0.5718
`hc_edgeconv_l8`	0.4292	0.5899	0.2467	0.5514
`mhc_edgeconv_l8`	**0.4360**	**0.5974**	0.2553	0.5631
`mhc_lite_edgeconv_l8`	0.4315	0.5927	0.2560	0.5621

For reference, the final GCN-family results at L8 are:

Table 7: Reference GCN-family results at L8

Model	OOD Kendall	OOD Spearman	Precision@10	NDCG@10
`gcn_l8`	0.3597	0.5055	0.2640	0.5632
`gcn_residual_l8`	0.3685	0.5138	0.2673	0.5543
`hc_gcn_l8`	0.4061	0.5576	**0.2867**	0.5669
`mhc_gcn_l8`	0.3964	0.5479	0.2780	0.5678
`mhc_lite_gcn_l8`	0.3945	0.5442	0.2820	0.5610

12.4 Interpretation

Observed behavior.

- EdgeConv alone is not enough:

$$\texttt{edgeconv_l8} = 0.3321 < \texttt{gcn_l8} = 0.3597$$

  on OOD Kendall.
- Adding a simple residual path changes the picture completely:

$$\texttt{edgeconv_residual_l8} = 0.4324$$

  which is above every GCN-family model on global OOD ranking.
- Adding HC/mHC/mHC-lite on top of EdgeConv gives only small deltas:

$$\texttt{mhc_edgeconv_l8} = 0.4360 \quad \text{vs.} \quad \texttt{edgeconv_residual_l8} = 0.4324$$

- On top-K metrics, the best GCN variants remain stronger:

$$\texttt{hc_gcn_l8} = 0.2867 \quad \text{vs.} \quad \texttt{edgeconv_residual_l8} = 0.2613$$

  on Precision@10.

Interpretation. The backbone matters more than initially hoped. On GCN, hyper-connections are the main source of the gain. On EdgeConv, the main source of the gain is the stronger convolution itself plus a standard residual path. The marginal utility of HC/mHC/mHC-lite becomes much smaller once the backbone is already strong.

Main conclusion. EdgeConv + residual is stronger than the final GCN family on global OOD ranking, but the comparison is not like-for-like in terms of model class and capacity. HC-GCN remains competitive if the target metric is top-K retrieval rather than global ranking.